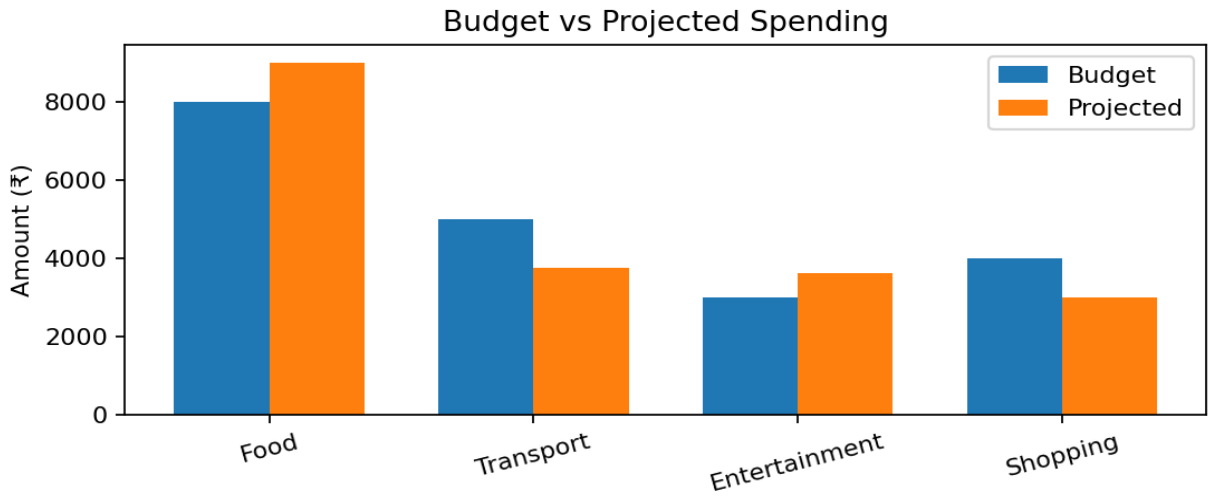
5.3 Expense Prediction

Overspending Prediction

Overspending Prediction compares current spending with a planned budget and estimates whether the user may exceed a category limit. The assistant can use the amount already spent, the remaining days, and the spending rate to produce an early warning. This gives users time to reduce unnecessary spending before the budget is exceeded.

Objectives

- Compare current spending with budget limits.
- Calculate the remaining budget.
- Estimate spending based on the current rate.
- Identify categories at risk of overspending.
- Provide an early warning.



Working Procedure

1. Read the planned category budget.
2. Record current spending and days passed.
3. Calculate the daily spending rate.
4. Estimate spending for the full month.
5. Compare the estimate with the budget and show a warning if needed.

Example Data Table

Category	Budget	Spent So Far	Projected Total	Prediction
Food	■8,000	■6,000	■9,000	Possible overspending
Transport	■5,000	■2,500	■3,750	Within budget

Category	Budget	Spent So Far	Projected Total	Prediction
Entertainment	■3,000	■2,400	■3,600	Possible overspending
Shopping	■4,000	■2,000	■3,000	Within budget

Python Implementation

```
budget = 8000
spent = 6000
days_passed = 20
days_in_month = 30

daily_rate = spent / days_passed
projected = daily_rate * days_in_month

print("Daily spending rate:", round(daily_rate, 2))
print("Projected monthly spending:", round(projected, 2))

if projected > budget:
print("ALERT: Possible overspending")
else:
print("Spending is projected to stay within budget.")
```

Benefits

Overspending prediction provides an early warning rather than waiting until the budget has already been exceeded. Users can then review optional spending and make adjustments.

Note: The values shown are illustrative examples for an academic project. A real application can use transaction records stored in a database.